Shistata Subedi

PhD Student · Computer Science · University of Oregon

shiss@uoregon.edu | shistuu.github.io | github.com/Shistuu | linkedin.com/in/shistatasubedi

Education

Degree	Institution	Years
Ph.D., Computer Science	University of Oregon, Eugene, OR	2024 – present
B.Tech., Computer Science	Vellore Institute of Technology, India	2019 – 2023

Publications

Smaran: Serving Authenticated Time-Travel Queries [Paper] [Code] *SOSP 2026*

- **Authors:** A. Nepal*, **S. Subedi***, S. Mishra*, S. Gupta, A. Kate, H. Meling (*equal contribution)
- Lets an untrusted server prove what a record looked like at a past time or across an interval, combining segment trees with KZG vector commitments for a constant number of fixed-size proofs.
- Applied to key transparency and Ethereum: 5.7× higher query throughput than CONIKS and OPTIKS, and up to 589× higher than Merkle and Verkle trees on 2.6M Ethereum blocks.

Research Projects

HotSwap: Workload-Adaptive Placement for Authenticated State [Results] *ACM SRC, 3rd place · SOSP 2026*

- Cuts the cost of keeping KZG witnesses up to date in authenticated storage when update traffic is highly skewed, without weakening what clients can verify.
- Evaluated on historical workloads, with about 3× less modeled witness-maintenance work.

Cry-for-Help: Escaping Metastable Failures *University of Oregon · 2026*

- Studies how distributed services can detect and recover from metastable overload without operator intervention.
- Built multi-node failure experiments with workload generation and resource throttling.

MiniSpanner: Sharded, Replicated Database [Code] *University of Oregon · 2025*

- Built on Apache ResilientDB, with PBFT inside each shard and 3PC / Proof-of-Execution coordinating transactions across shard leaders.

Experience

Software Engineer, Leapfrog Technology *Kathmandu, Nepal · 2023 – 2024*

- Designed and optimized backend services and REST APIs in Java and Micronaut, improving response times by 30%.
- Built features for Aspen Biosciences' drug-discovery platform, covering experiment tracking, chemistry workflows, and lab inventory.
- Streamlined delivery workflows, cutting project timelines by 20%, and led annotation work that raised labeling accuracy by 25%.

Teaching Assistant, University of Oregon *Eugene, OR · 2024 – 2025*

- Taught web programming lectures and ran hands-on debugging labs for 140+ students.

Honors & Awards

3rd Place (Graduate), ACM Student Research Competition, SOSP — HotSwap *2026*
Lokey Award for Graduate Excellence, University of Oregon *2024 – 2025*
Frank Vignola Microgrant, Eugene–Kathmandu Sister City Association *2025*
COMPEX Full Undergraduate Scholarship, Embassy of India *2019 – 2023*

Talks & Service

- **Presentations:** SRC poster, SOSP 2026 · Doctoral Symposium, Middleware 2025 · Poster, USENIX OSDI 2025
- **Service:** Reviewer, IEEE TDSC · Shadow Reviewer, ACM IMC 2025 · Artifact Evaluation Committee, Middleware 2025

Skills

- **Languages:** C++, Go, Rust, Python, Java, JavaScript, TypeScript
- **Security & Cryptography:** Authenticated data structures, KZG polynomial and vector commitments (BLS12-381), Merkle and Verkle trees, zero-knowledge proofs, range proofs, proof aggregation, FFT/NTT-based protocols, verifiable computation, key transparency
- **Distributed Systems:** Byzantine fault tolerance (PBFT), consensus and replication, sharding, distributed transactions (3PC), blockchain protocols, storage systems, overload and failure recovery
- **Tools:** Linux, Git, Docker, CloudLab, Ethereum/Geth, PebbleDB, MySQL, MongoDB, Micronaut